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SUGGESTED PRESENTATION OPENING

Opening: why LandVerifyCM matters

Good morning. Land is often a family's lifetime investment. Yet in Cameroon, forged titles, duplicate records and unclear ownership can turn it into a dispute. LandVerifyCM makes an early check faster and more transparent.

OUR ANSWER A centralized platform for citizens to search land records and authorized administrators to manage parcel information.

PURPOSE

The project in one view

- **Public confidence** — search by title, parcel ID or owner; view status, location and history.
- **Fraud warning** — classify records using title uniqueness and GPS proximity.
- **Accountability** — protect record management by role and trace important changes.
- **Accessible design** — connect web and mobile interfaces to one API with English/French support.

COORDINATION

My leadership role

As project leader, I aligned three workstreams: Nkam Titcha on frontend/mobile, Kemgang Leprince on backend services, and my database work. I coordinated shared data contracts so every layer used one trusted structure.

DATABASE MANAGER

My responsibility: trustworthy data

I designed the PostgreSQL structure through Prisma, prepared migrations and seed data, and connected it to verification and administration. The database preserves identity, history, evidence and responsibility.

DATA MODEL	WHAT IT CONTRIBUTES
User	Accounts, ADMIN/USER role, active state and secure password hash.
LandParcel	Unique title, owner, quarter, area, GPS, land use and verification status.
OwnershipRecord	Chronological chain: original ownership, purchase, inheritance, donation or court order.
LandDocument	Supporting PDF/image metadata linked to its parcel.
AuditLog	Who performed create, update, deactivation or ownership transfer, and what changed.

IMPLEMENTED SAFEGUARDS

How the data stays reliable

- **Integrity** — UUID keys, foreign keys and unique title numbers prevent ambiguity.
- **Traceability** — documents, ownership and audit events link to the correct parcel.
- **Performance** — indexes accelerate quarter, status, active-record and relation lookups.
- **Safe lifecycle** — deactivation retains records; related evidence follows controlled cascade rules.

TECHNICAL CHOICES

Languages and technologies—and why

CHOICE	USED FOR	WHY IT FITS
PostgreSQL + SQL	Storage & migrations	Strong constraints, joins and durable structured records.
Prisma ORM	Schema and access	Type-safe queries, relations and repeatable migrations.
TypeScript	Seed and integration	Shared types reduce API-to-database mistakes.

END-TO-END DATA FLOW

From a search to a trusted result

01

CAPTURE

Admin submits parcel, ownership and evidence.

02

CHECK

API validates values and compares title/GPS.

03

STORE

Prisma writes linked PostgreSQL records.

04

TRACE

Audit history records responsible actions.

PRACTICAL VALUE

How this can help Cameroonian people

- **Before payment** — buyers can check beyond the seller's paper documents.
- **Earlier fraud detection** — duplicate titles or overlapping GPS locations trigger warnings.
- **Clearer ownership** — families and professionals can follow transfers over time.
- **Better access** — search, maps and bilingual interfaces simplify land information.
- **Accountable records** — audit logs connect sensitive actions to an administrator.

RESPONSIBLE SCOPE LandVerifyCM supports due diligence and informed decisions; formal ownership remains subject to verification by Cameroon's competent land authorities.

SUGGESTED CLOSING

Conclusion

To conclude, LandVerifyCM applies software engineering to a real national concern. The interfaces make verification approachable, the backend applies rules, and the database preserves evidence and history. As project leader and database manager, I connected these parts around one source of truth.

For Cameroonians, this means fewer blind decisions, earlier warnings and clearer information before committing hard-earned money. The platform does not replace public authorities; it provides a stronger first check and a transparent path to verification.

Thank you.